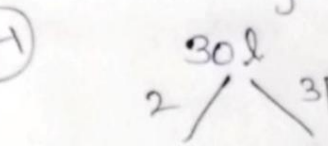
(99)

(ii) half of mixture replaced total 3 times from the Question, here replacing nothing but total quantity of mixture is same throughout in 3 more replacements.

(M-1)



(2+3)P = 30 l
1P = 6 l.

after 1st Replacement → 6
after 2nd → 3
after 3rd → 1.5

9 + (6+9) = 24
12 + (3+12) = 27
13.5 + (1.5+13.5) = 28.5

1.5 : 28.5 = 15 : 285

$M_f : W_f = 1 : 19$

(M-2) - Milk quantity is getting half in every transaction. So, we can find out M_f directly

$M_f = M_i \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 12 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
$= 1.5$ ltrs.

W_f = Total (doesn't change) $- M_f$
$= 30 - 1.5 = 28.5$

$M_f : W_f = 1.5 : 28.5 = 1 : 19$

(M-3) from (M-2) we can create a solution for any total mixture quantity of ratio only given [means 30 ltrs is not given in the question]. $M : W = 2 : 3 \Rightarrow M_i = \frac{2}{5}, W_i = \frac{3}{5}$

$M_f = M_i \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{2}{5} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{20}$

means milk finally 1 part of 20 total parts then the remaining parts are water. only.

$M_f : W_f = 1 : (20-1) = 1 : 19$

(M-4) formula:- from M-3 we create a formula like.

$M_f = M_i \left(1 - \frac{1}{2}\right)^3 = M_i \left(1 - \text{Replacing quantity}\right)^{n \text{ times}}$.

Ex:- If $\frac{1}{3}^{rd}$ quantity is replacement in the above question

$M_f = \frac{2}{5}\left(1 - \frac{1}{3}\right)^3 = \frac{2}{5} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{135}$

$M_f : W_f = 8 : (135-8) = 8 : 127$.

M_f in ltrs $= M_i$ (in ltrs) $\left(1 - \text{Replacement quantity}\right)^{n \text{ times}}$

Ex:8:- When 2 mixtures of M & W containing them in the ratio of 2:3 & 3:4 are mixed.
(i) In equal volumes (1:1) (ii) In the ratio of 2:3 then the resultant ratios are?

	Milk	Water	
Mix:1 (2:3) →	$\frac{2}{5}$	$\frac{3}{5}$	×1
Mix:2 (3:4) →	$\frac{3}{7}$	$\frac{4}{7}$	×1
	M	W	

(i) $1:1 \Rightarrow M_1:W_1$

$\frac{2}{5}\times1+\frac{3}{7}\times1 : \frac{3}{5}\times1+\frac{4}{7}\times1$

$\frac{14+15}{35} : \frac{21+20}{35}$

(ii) 2:3

(i) $\boxed{29:41}$

(ii)
$$\begin{bmatrix} \frac{2}{5} & \frac{3}{5} \end{bmatrix} \times 2$$
$$\begin{bmatrix} \frac{3}{7} & \frac{4}{7} \end{bmatrix} \times 3$$

$\Rightarrow M_2:W_2 = \frac{2}{5}\times2+\frac{3}{7}\times3 : \frac{3}{5}\times2+\frac{4}{7}\times3$

$= \frac{14\times2+15\times3}{35} : \frac{21\times2+20\times3}{35}$

(ii) $\boxed{M_2:W_2 = 73:102}$

↳ Note:- When ratio is devided / multiplied with any number (to all individual parts) then the ratio doesn't change.

↳ When a ratio is added / subtracted by any value to all its individual quantities no. of parts then the resultant ratio changes.

Averages & Ages [Vedor India Pvt. Ltd]

↳ By Sunil Guptha Yavramalla

What is Average?

↳ Equal distribution among all the observations is called average.

↳ To distribute equally total sum is devided by total no. of observations.

$$\Rightarrow \boxed{\text{Average} = \frac{\text{Sum of Observations}}{\text{No. of Observations}}}$$

$$A = \frac{S}{N} \Rightarrow N = \frac{S}{A} \Rightarrow S = N \times A.$$

↳ When all observations are multiplied / devided by a constant number then their Avg is Original Avg multiplied/ devided by same number.

↳ When all observations are added/ substracted by a constant number then resultant Avg is also added/ substracted by same number to original Average.

Averages of:-

1) Average of 1st 'n' natural Numbers $= \frac{\text{Sum of obs}}{\text{no. of obs}}$

$$= \frac{\text{Sum of 1st n natural No.s}}{n} = \frac{n(n+1)/2}{n}$$

$$= \left[\frac{n+1}{2}\right]$$

2) Average Squares of 1st 'n' natural No.s :-

$$= \frac{n(n+1)(2n+1)}{6} \Big/ n = \frac{(n+1)(2n+1)}{6}$$

3) Average of cubes of 1st 'n' natural Numbers.

$$= \frac{\left[\frac{n(n+1)}{2}\right]^2}{n} = \frac{n^2(n+1)^2}{4n} = \frac{n(n+1)^2}{4}$$

4) Average of 1st 'n' odd numbers = n

$$= \frac{\text{Sum of 1st 'n' odd no.s}}{\text{no. of obs}} = \frac{n^2}{n} = n$$

5) Average of 1st 'n' even numbers = $(n+1)$

$$= \frac{\text{Sum of 1st 'n' even NO.S}}{n} = \frac{n(n+1)}{n} = (n+1)$$

Ex:-1:- What is Avg of 1, 3, 5, 7, 9, 11, 13 ?

Ans:- M-1:- $\text{Avg} = \frac{1+3+5+7+9+11+13}{7} = \frac{49}{7} = 7$

M-2:- Given 7 first odd numbers then according to point (4) Avg is also '7'.

M-3:- Whenever the given all observations are in A.P (common diff is same), then Avg of all observations is Avg of 1st & Last observations.

$$\text{Avg} = \frac{\text{1st observation} + \text{Last observation}}{2}$$

$$= \frac{1+13}{2} = \frac{14}{2} = 7.$$

M-4:- Whenever the given all observations are in A.P and the no. of observations are odd number of observations, then the Average is exactly middle term

1, 3, 5, (7), 9, 11, 13 → 7 observations

↓

Middle one is average.